DSEH001 · DIGITAL TRANSFORMATION IN HR SYSTEMS

01

UNIT 1 READER

Why HR Is Going Digital

Drivers, maturity and the ethics of adoption

Weeks 1 to 3 | Exam preparation | 10 readings explained

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About this reader

How to use it. This reader is a companion to the set readings, not a replacement for them. For each reading it explains the argument in plain language, lists the ideas you must hold, shows how to apply them, flags common confusions, and gives questions to think with as you prepare for the end-term examination. Read the original first, then use this reader to check your understanding. Folder names refer to the course Readings folder.

Unit 1 asks a deceptively simple question: why is human resource management being rebuilt around digital technology, and how would you know how far an organization has travelled? The readings answer at three levels. At the level of the economy, Schwab argues that a new industrial era is changing how value is created. At the level of the firm, Westerman, Bonnet and McAfee show that technology alone does not create advantage; leadership capability decides who gains. At the level of the HR function, Stone and colleagues and Kavanagh and Johnson describe what technology has done to HR processes, and what it has failed to do.

The second half of the unit turns from drivers to diagnosis. A maturity model gives you a ladder against which to place an organization, and the unit contrasts two ladders that are often confused. The unit closes with a warning from Angrave and colleagues and from Manuti and de Palma: digital HR can fail quietly, through weak analytical capability or through a narrow view of what workers need, and it can fail loudly, through a breach of privacy law. Read the unit as a single argument: drivers create pressure, maturity models measure readiness, and ethics and law set the limits.

READING MAP

Week	Set readings	Folder	Exam-edition lectures
1	Schwab (2016) Introduction and Part I; Stone et al. (2015)	Unit-1 Core	L2
2	Westerman et al. (2014) Ch 1 to 2 and the maturity framework; Scully-Russ and Torraco (2020) pp. 73 to 77; Deloitte Human Capital Trends (selected). Frameworks: Kavanagh maturity stages; Bersin (2012)	Unit-1 Core, Unit-1 Supp., Unit-2 Core, Unit-4 Supp.	L1, L2
3	Kavanagh and Johnson (2018) Ch 2 and 14; Angrave et al. (2016); Manuti and de Palma (2018) Ch 3 and 6; DPDP Act 2023 orientation note	Unit-2 Core, Unit-3 Core, Unit-4 Supp.	L1 to L3

The readings, one by one

Week 1 Core

Schwab, K. (2016). *The Fourth Industrial Revolution*. World Economic Forum. Introduction and Part I.

THE ARGUMENT IN BRIEF

Schwab places today's change in a sequence of industrial revolutions: mechanization, mass production, and the computer and internet era. He argues that the present period is different in kind, not only in degree, because digital, physical and biological technologies are fusing. He gives three reasons why it counts as a revolution: the speed of change, its breadth and depth across every industry, and its impact on whole systems, including firms, countries and society. His central concern is that institutions, including employers, are adapting far more slowly than the technology.

IDEAS TO HOLD

- Velocity: change now moves at an exponential rather than a linear pace.
- Breadth and depth: many technologies combine, so change reaches business models, not only tools.
- Systems impact: whole systems of production, management and governance are reshaped together.
- The gap between technological speed and institutional adaptation is the central management risk.

APPLY IT

Schwab's gap between technological speed and institutional adaptation describes most HR functions: the systems exist, but the organization has not adapted to use them. Use it to argue that the problem is organizational, not technical.

QUESTIONS TO THINK WITH

- Which of Schwab's three features does an HR department feel first, and why?
- Is the gap Schwab describes larger for a retailer with a large frontline in smaller towns than for a software firm? Argue from one example.

Week 1 Core

Stone, D. L., Deadrick, D. L., Lukaszewski, K. M., & Johnson, R. (2015). The influence of technology on the future of human resource management. *Human Resource Management Review*, 25(2), 216 to 231.

THE ARGUMENT IN BRIEF

Stone and colleagues review how technology has changed the main HR processes: acquiring talent, developing it, managing performance and rewarding people. They show that technology has changed how HR collects, stores, uses and shares information, and that it promised lower cost and better decisions. Their contribution is to ask whether those promises were met. They find the evidence on effectiveness thin, and they list unintended consequences: privacy concerns, less personal contact between HR and employees, and systems that can screen out good candidates. They close with a research agenda that asks whether technology helps organizations reach HR goals, not only efficiency goals.

IDEAS TO HOLD

- Technology changed the information flows of HR before it changed HR's purpose.
- Efficiency gains are better documented than effectiveness gains.
- Unintended consequences include privacy risk, depersonalization and unfair screening.
- The right test is whether HR goals (attracting, motivating, retaining) are met, not whether a system is live.

APPLY IT

A system can be live and widely used and still fail the effectiveness test. Ask of every HR system: which HR goal does it serve, and what evidence shows it serves it?

EASY TO CONFUSE

Efficiency (doing HR tasks cheaply) and effectiveness (reaching HR goals) are different claims. Many student answers prove the first and assert the second.

QUESTIONS TO THINK WITH

- Pick one HR process in an organization you know. State one efficiency claim and one effectiveness claim for it, and the evidence each would need.

Week 2 Core

Westerman, G., Bonnet, D., & McAfee, A. (2014). *Leading Digital: Turning Technology into Business Transformation*. Harvard Business Review Press. Chapters 1 and 2.

THE ARGUMENT IN BRIEF

Westerman and colleagues studied large firms outside the technology sector and asked why some gain from digital technology and others do not. Their answer is that advantage depends on two separate capabilities. Digital capability is what the firm does with technology: how it engages customers, runs operations and builds new business models. Leadership capability is how the firm steers change: vision, engagement, governance and the link between IT and the business. Crossing the two gives four kinds of firm: Beginners (low on both), Fashionistas (many digital initiatives, weak leadership), Conservatives (strong governance, cautious digital activity) and Digital Masters (strong on both). The opening chapters set out this framework and begin with the customer experience as the first place digital change becomes visible.

IDEAS TO HOLD

- Two dimensions: what you do with technology, and how you lead the change.
- Fashionistas buy many tools without coordination; the result is fragmentation, not transformation.
- Conservatives govern well but move slowly; caution can become its own risk.
- Digital Masters combine both capabilities, which is why buying technology alone rarely creates advantage.

> **APPLY IT**
>
> Overlapping systems bought by different departments (two HRIS, two learning platforms) are a classic Fashionista signal. The remedy is governance, not another product.

> **EASY TO CONFUSE**
>
> This quadrant is about the whole firm's digital strategy. It is not an HR maturity ladder, and it does not have stages that must be climbed in order.

QUESTIONS TO THINK WITH

- Which quadrant fits a typical large Indian firm's HR function? What evidence would you need to place it?
- Can an HR function be a Digital Master inside a Conservative firm? What would limit it?

Week 2 **Core**

Scully-Russ, E., & Torraco, R. J. (2020). The changing nature and organization of work: An integrative review of the literature. *Human Resource Development Review*, 19(1), 66 to 93. Week 2 pages: 73 to 77.

THE ARGUMENT IN BRIEF

This integrative review gathers research on how work itself is changing, and the Week 2 pages set out the antecedents of that change: technology, finance, worker behavior and institutional shifts. The authors treat technology as one driver among several, not the only one. Their key point for HR is that these drivers interact: technology changes tasks, which changes jobs, which changes the employment relationship and the skills workers need. The article returns in Weeks 5, 10 and 11, so read the Week 2 pages as the foundation for a longer argument.

IDEAS TO HOLD

- Work is changing through several interacting drivers, not technology alone.
- Change flows from tasks to jobs to the employment relationship.
- HR development has a role in helping workers adjust, not only in deploying systems.

APPLY IT

Avoid single-cause explanations of HR change. A driver map that links technology to workforce structure (frontline share, location, contract type) is stronger than a list of tools.

QUESTIONS TO THINK WITH

- Name two drivers other than technology that are changing Indian workforces, and the HR process each affects.

Week 2 Framework

Kavanagh, M. J., & Johnson, R. D. (Eds.). (2018). *Human Resource Information Systems: Basics, Applications, and Future Directions* (4th ed.). Sage. Used in the course as Kavanagh's maturity model.

THE ARGUMENT IN BRIEF

The course uses a five stage maturity ladder for the whole HR technology function, referred to as Kavanagh's maturity model: Level 1 is fully manual or paper based and Level 5 is intelligent and optimized, with spreadsheets, point solutions and integrated systems as the steps between. The logic of the ladder is cumulative. Each stage builds the data and process discipline the next one depends on, so an organization cannot reliably jump to analytics or AI while its core records are still on paper or in disconnected spreadsheets.

IDEAS TO HOLD

- The ladder describes the whole HR technology function, process by process.
- Stages are cumulative; skipping a stage leaves the next one without a foundation.
- The shape of the profile across processes is the diagnosis, not the average score.
- No organization should be assumed to sit at one stage across all processes.

APPLY IT

Profile maturity process by process: recruitment may sit at one stage, payroll at another. Name the highest and lowest processes and explain what the gap means for sequencing.

EASY TO CONFUSE

Do not confuse this ladder with Bersin's levels, which measure analytics capability only. Both ladders warn against skipping, so 'one allows skipping and the other does not' is not the difference.

QUESTIONS TO THINK WITH

- Why is an average maturity score a weak summary for a Board?
- Which low-maturity process would block a predictive analytics project most directly?

Week 2 Framework

Bersin, J. (2012). *Talent analytics maturity model.* Bersin by Deloitte.

THE ARGUMENT IN BRIEF

Bersin's model describes how far an organization's people analytics has developed, in four levels. Level 1 is operational, reactive reporting: counting and describing what happened. Level 2 is advanced, proactive reporting: benchmarking and trend dashboards that help managers act. Level 3 is advanced analytics, the strategic level: statistical models that explain why outcomes occur and link people data to business results. Level 4 is predictive analytics: forecasting outcomes and planning interventions. Bersin's research presented the levels as a progression that organizations must climb in order, and found that very few organizations, about 4 percent, had reached the predictive level.

IDEAS TO HOLD

- Four levels: Reactive reporting, Proactive reporting, Strategic analytics, Predictive analytics.
- The model covers analytics capability only, not the whole HR technology stack.
- Moving up depends on clean data, analytical skills and sponsorship.
- Owning a predictive tool does not place an organization at the predictive level.

APPLY IT

An organization that owns a flight-risk tool is not thereby at Bersin's predictive level; check whether reliable reporting and joined data exist underneath it.

EASY TO CONFUSE

Bersin's levels and Kavanagh's stages are two different ladders. Placing an organization on one does not place it on the other.

QUESTIONS TO THINK WITH

- Can an organization be Level 4 on Bersin and low on the HR technology ladder? What would that look like in practice?

Week 3 Core

Kavanagh, M. J., & Johnson, R. D. (2018). Chapters 2 and 14.

THE ARGUMENT IN BRIEF

The Week 3 chapters build the data foundation for everything that follows. Chapter 2 explains how data become information and knowledge, and introduces databases, data warehouses, business intelligence and data mining: the layers that sit between an HR transaction and a manager's dashboard. Chapter 14 turns to HR metrics and workforce analytics: how measures such as turnover, time to fill and cost per hire are defined, how they are benchmarked, and how they can be connected to business outcomes. The authors stress that a metric is only useful when its definition is agreed and its data are reliable.

IDEAS TO HOLD

- Data, information and knowledge are different; a warehouse and a BI layer turn the first into the third.
- A single source of truth is a design choice, not a software feature.
- Every metric needs an agreed definition, a reliable data source and a clear user.
- Workforce analytics adds value when it links HR measures to business results.

> **APPLY IT**
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> Before quoting any HR figure, write its definition: numerator, denominator, period and data source. This habit prevents most errors in HR reporting.

> **QUESTIONS TO THINK WITH**
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> - Write a full definition for attrition rate, as Chapter 14 would require.

Week 3 **Core**

Angrave, D., Charlwood, A., Kirkpatrick, I., Lawrence, M., & Stuart, M. (2016). HR and analytics: Why HR is set to fail the big data challenge. *Human Resource Management Journal*, 26(1), 1 to 11. Set again in Week 8.

THE ARGUMENT IN BRIEF

Angrave and colleagues argue that HR analytics was being oversold. Vendors offered dashboards and data tools, but HR functions often lacked the analytical skill and the theoretical understanding of people at work needed to turn data into insight. Without that understanding, analytics tends to count what is easy to count and to confuse correlation with cause. The authors call for HR to build real analytical capability, to ground models in theory about how people behave, and to use stronger designs such as experiments where possible. Otherwise, they warn, analytics will be captured by others, and HR will lose influence.

IDEAS TO HOLD

- Tools are not capability; skill and theory are the scarce resources.
- Data driven without theory leads to counting the easy things and misreading correlations.
- Stronger evidence comes from better designs, including experiments, not bigger dashboards.
- If HR cannot do analytics well, other functions will do it for HR.

APPLY IT

This reading underpins later lectures on audits and dashboards: a model's heavy weight on a variable is not proof that the variable causes attrition.

QUESTIONS TO THINK WITH

- Name one HR measure that is easy to count but hard to interpret. What theory would you need to read it well?

Week 3 **Core**

Manuti, A., & de Palma, P. D. (2018). *Digital HR: A Critical Management Approach to the Digitilization of Organizations*. Palgrave Macmillan. Chapters 3 and 6 (2018 edition, filed in Unit-2 Core).

THE ARGUMENT IN BRIEF

Manuti and de Palma take a critical management view of digital HR. Chapter 3 examines the cognitive technology revolution and what it does to HR practices; Chapter 6 is the book's conclusion. Drawing on work psychology, they ask how cognitive technologies change workers' identity and the meaning they find in work, not only how they change HR processes. Their evidence comes from interviews and case material from HR managers in large organizations. The book's recurring message is that digital tools must be framed within the organization's culture and its view of people. A system introduced without attention to identity, trust and meaning can be technically successful and still damage the employment relationship.

IDEAS TO HOLD

- Digital change reshapes worker identity and the meaning of work.
- HR's role shifts toward framing technology within culture.
- Case evidence shows success depends on how tools fit people, not only on features.
- A critical approach asks who gains and who bears the cost of digitalization.

APPLY IT

A rollout that treats frontline staff as users to be converted, rather than workers whose roles are changing, meets resistance the dashboards cannot explain.

QUESTIONS TO THINK WITH

- Which group's sense of work would change most under a mobile-first HR model, and what should HR do about it?

Week 3 **Statute**

Government of India. (2023). *The Digital Personal Data Protection Act, 2023* (Act No. 22 of 2023).

THE ARGUMENT IN BRIEF

The DPDP Act governs digital personal data in India. It calls the person whose data is processed the Data Principal and the organization that decides why and how data are processed the Data Fiduciary. Processing generally needs a lawful basis: consent that is free, specific, informed, unconditional and unambiguous, given after a clear notice, or one of the legitimate uses listed in Section 7, which include purposes connected with employment. Data Principals have rights to information, correction, erasure and grievance redress. Fiduciaries must keep data accurate and secure, report breaches, and delete data once the purpose is served. The government can restrict transfers to notified countries, and a Data Protection Board can impose large penalties.

IDEAS TO HOLD

- Two roles: Data Principal (the employee) and Data Fiduciary (the employer).
- Lawful basis: consent with notice, or a listed legitimate use such as certain employment purposes.
- Duties include purpose limitation, accuracy, security, breach reporting and deletion.
- Cross border transfer is allowed unless the destination is restricted by notification.

APPLY IT

Many HR systems store data abroad. Flag compliance gaps before building: a gap found early costs a memo; found late, it costs a crisis.

EASY TO CONFUSE

The compliance first audit (checking what exists against the law) is not the same as Privacy by Design (building protection into a new system). You will meet the second in Unit 2.

QUESTIONS TO THINK WITH

- Does a flight-risk model fit within the employment legitimate use, or does it need consent? Argue both sides.

Unit 1 frameworks side by side

Framework	What it measures or claims	Commonly confused with
Kavanagh maturity stages	Whole HR technology function, five stages; stages build on each other	Bersin levels
Bersin analytics levels	Analytics capability only: Reactive, Proactive, Strategic, Predictive	Kavanagh stages
Digital inclusion (course framing)	Access, Skills, Use; fixing one layer leaves the other two in place	ADKAR Ability
Compliance first (DPDP 2023, SS2020)	Find legal gaps before building; early is cheap, late is a crisis	Privacy by Design
Kotter Step 1	Create a sense of urgency grounded in evidence	Lewin's unfreeze phase

TWO MORE IDEAS YOU WILL NEED

Digital inclusion: Access, Skills, Use. The course frames the digital divide in three layers. Access is having a usable device and connection. Skills is being able to operate the system. Use is actually choosing to use it for real tasks. The layers are linked: training (a Skills fix) does nothing for someone whose only device is a feature phone (an Access barrier). Segment a workforce by device access, digital literacy and location before recommending a fix.

Kotter's first step: urgency. Kotter's eight step model (1996) begins by establishing a sense of urgency. Urgency is not alarm; it is a shared, evidence based case that the status quo is more dangerous than change. A maturity diagnosis is, in effect, an organization's Step 1 document. A widely repeated claim that 70 percent of change efforts fail is not in Kotter's books and is poorly supported, so do not cite it.

Key terms

Data Fiduciary	Under the DPDP Act, the organization that decides the purpose and means of processing personal data.
Data Principal	The person to whom personal data relate; in HR, the employee.
Digital Master	Westerman's term for a firm strong in both digital and leadership capability.
Effectiveness	Whether HR goals are achieved, as distinct from efficiency, which is cost per task.
Maturity model	A ladder of stages used to place an organization and plan its next step.
Point solution	A system that handles one process and does not share data with others.

References

Angrave, D., Charlwood, A., Kirkpatrick, I., Lawrence, M., & Stuart, M. (2016). HR and analytics: Why HR is set to fail the big data challenge. *Human Resource Management Journal*, 26(1), 1 to 11.

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Westerman, G., Bonnet, D., & McAfee, A. (2014). *Leading digital*. Harvard Business Review Press.

Deloitte's Global Human Capital Trends 2026 (Unit-4 Supplementary) is also set for Week 2. Read it for current context and note which trends Indian organizations are already facing; this reader does not summarize it.